THE ASPECT RATIO OF THE TWIN DRAGON IS $1/\varphi$

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Abstract. We show that the geometric aspect ratio of the Twin Dragon — defined via the Gaussian integer $1+i$ — equals $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio, and that its major axis makes angle $-\arctan\varphi$ with the horizontal. More generally, for every equal-weight IFS $\{f_k(z) = a(z+t_k)\}$ with centred translations the aspect ratio depends only on a and a single anisotropy parameter $\kappa = |\sum t_k^2|/\sum |t_k|^2 \in [0,1]$. When $a = 1/\lambda$ for a Gaussian integer λ, the aspect ratio lies in a quadratic field determined by $\mathbf{N}(\lambda^2-1)$. Every metallic ratio $\mu_m = (m+\sqrt{m^2+4})/2$ $(m \geq 1)$ arises as the reciprocal aspect ratio of a plane-filling tile over $\mathbb{Z}[i]$; moreover, $\mathbb{Z}[i]$ is the unique imaginary quadratic ring where collinear digits can produce metallic-ratio aspect ratios.

1. Introduction

The Twin Dragon is a classical plane-filling fractal [4] governed by the Gaussian integer $1+i$: its two defining contractions share the linear part $(1-i)/2$, and their translations are symmetric about the origin. Despite this simple, purely arithmetic construction, the attractor has a non-trivial shape that is not immediately obvious from the definition. Visually the Twin Dragon looks somewhat elongated — neither a square nor a thin sliver — which raises the question of its precise aspect ratio.

A natural quantitative measure of shape for a planar set is the *geometric aspect ratio*: the ratio of standard deviations along the principal axes of its area distribution, or equivalently the square root of the ratio of the eigenvalues of the second-moment (covariance) matrix M of the uniform measure on the attractor. This definition gives the expected answers for elementary shapes: an ellipse with semi-axes $\alpha \geq \beta > 0$ has aspect ratio β/α (eigenvalues $\alpha^2/4$ and $\beta^2/4$, so the definition directly recovers the ratio of the semi-axes); a $p \times q$ rectangle has aspect ratio p/q (eigenvalues $p^2/12$ and $q^2/12$); and a square has aspect ratio 1.

The approach of characterising IFS attractors through moments of their invariant measures was introduced by Vrscay and Roehrig [10] and systematically developed in [11]. The key observation is that the self-similar measure satisfies a linear fixed-point equation in its moments, which can be solved exactly. When the open set condition holds, Hutchinson's theorem [6] guarantees that the self-similar measure coincides with the normalised Hausdorff measure $\mathcal{H}^s|_A$, so the covariance matrix captures the actual geometry of the attractor at whatever dimension s it carries. In the special case where the similarity dimension s equals the ambient dimension, $\mathcal{H}^s$ reduces to Lebesgue area.

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We apply this method to the Twin Dragon. The covariance matrix turns out to be $M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$, whose eigenvalues involve $\sqrt{5}$, leading to the aspect ratio $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The appearance of φ in a fractal with no pentagonal geometry is unexpected. Although the method of moments for IFS goes back to [10], the closed-form aspect ratio formula for equal-weight IFS with centred translations (Theorem 1 below) and the characterisation of its quadratic field for Gaussian-integer parameters (Corollary 2) are, to our knowledge, new.

Discovery. The result was first found empirically using the IFStile software [9], which implements the moment invariants of [10, 11] as part of its algebraic IFS search engine. Among the similarity invariants reported by IFStile for the Twin Dragon, the aspect ratio $1/\varphi$ appeared as a numerical value; the present paper supplies the exact proof and its generalisation to all equal-weight IFS $\{f_k(z) = a(z+t_k)\}$ with centred translations. This is an instance of *experimental mathematics* [3]: a computer experiment suggests a precise conjecture, which is then confirmed by a rigorous proof.

The paper is organised as follows. Section 2 establishes a closed-form aspect ratio formula (Theorem 1) for equal-weight IFS $\{f_k(z) = a(z + t_k)\}$ with centred translations and classifies the quadratic field of the aspect ratio (Corollary 2). Section 3 applies the formula to a two-parameter family of plane-filling tiles parameterised by $x^2-\ell x+N$ (Proposition 4). Section 4 specialises to tiles over $\mathbb{Z}[i]$, proves that every metallic ratio μ_m arises as the reciprocal aspect ratio of such a tile (Theorem 7), and shows that $\mathbb{Z}[i]$ is the unique imaginary quadratic ring where collinear tiles achieve metallic-ratio aspect ratios (Corollary 9). Section 5 applies the formula to the Twin Dragon and proves the main result.

2. The aspect ratio formula

An *iterated function system* (IFS) is a finite collection of contractions $f_1, \dots, f_N$ on a complete metric space. By Hutchinson's theorem [6] there exists a unique non-empty compact set A, the *attractor*, satisfying $A = f_1(A) \cup \cdots \cup f_N(A)$. For background see [5]; for self-affine digit tiles in particular, see [7, 8].

Consider an equal-weight IFS of $N \geq 2$ contractions on $\mathbb{C}$, each of the form

$$f_k(z) = a(s_k z + t_k) \qquad \text{or} \qquad f_k(z) = a(s_k \overline{z} + t_k),$$

where $a \in \mathbb{C}$, $|a| < 1$, the rotations satisfy $s_k \in S^1$, and $t_1, \dots, t_N \in \mathbb{C}$ satisfy $\sum_{k=1}^N t_k = 0$. The condition $\sum t_k = 0$ is no loss of generality: shifting all translations by a constant merely translates the attractor without changing the covariance matrix. Define the *digit variance* $\sigma^2 := \frac{1}{N}\sum_k |t_k|^2 > 0$, the *algebraic variance* $\tau := \frac{1}{N}\sum_k t_k^2 \in \mathbb{C}$, and the *digit anisotropy*

$$\kappa := \frac{|\tau|}{\sigma^2} = \frac{|\sum t_k^2|}{\sum |t_k|^2} \in [0, 1].$$

The bound $\kappa \leq 1$ follows from the triangle inequality $|\sum t_k^2| \leq \sum |t_k^2| = \sum |t_k|^2$. When all t_k are collinear ($t_k = e^{i\theta} r_k$ with $r_k \in \mathbb{R}$), one has $\kappa = 1$; vertices of a regular N-gon ($N \geq 3$) give $\kappa = 0$. The Twin Dragon is the case $N = 2$, $a = (1 - i)/2$, $s_k = 1$, $t_k = \pm 1$ ($\sigma^2 = 1$, $\kappa = 1$), studied in Section 5.

Theorem 1. *With the notation above, let I and J denote the sets of orientation-preserving and -reversing indices, and define*

$$u = \frac{1}{N}\sum_{k\in I} s_k^2, \qquad v = \frac{1}{N}\sum_{k\in J} s_k^2.$$

Then the second moment $\omega := E_\mu[Z^2]$ is

$$\omega = \frac{a^2\big(\tau + \overline{a^2}(v\,\overline{\tau} - \bar{u}\,\tau)\big)}{|1-a^2u|^2 - |a|^4|v|^2}, \tag{1}$$

the aspect ratio is $\mathrm{AR}^2 = (1-\rho)/(1+\rho)$, where

$$\rho = \frac{(1-|a|^2)\,|\omega|}{|a|^2\,\sigma^2}, \tag{2}$$

and the major axis of the attractor makes angle

$$\psi = \tfrac{1}{2}\arg\omega \tag{3}$$

with the positive x-axis. When $v = 0$ (all maps orientation-preserving), (1) reduces to $\omega = a^2\tau/(1-a^2u)$, and

$$\rho = \frac{(1-|a|^2)\,|\tau|}{|1-a^2u|\,\sigma^2}. \tag{4}$$

When additionally all $s_k = 1$, we have $u = 1$, and with $\kappa = |\tau|/\sigma^2 \in [0,1]$:

$$\mathrm{AR}^2 = \frac{|1-a^2| - (1-|a|^2)\,\kappa}{|1-a^2| + (1-|a|^2)\,\kappa}, \tag{5}$$

In this case ($u = 1$, $v = 0$):

(a) AR *depends on the translations only through κ.*
(b) *For fixed a,* AR *is strictly decreasing in κ: $\kappa = 0$ (isotropic digit set) gives* $\mathrm{AR} = 1$, *and $\kappa = 1$ (collinear translations) gives the most elongated attractor.*
(c) $\mathrm{AR} = 0$ *if and only if $a \in \mathbb{R}$ and $\kappa = 1$.*

For collinear translations ($\kappa = 1$, $\tau > 0$ after aligning the digit axis with $\mathbb{R}$),

$$\tan 2\psi = \frac{\mathrm{Im}(a^2)}{\mathrm{Re}(a^2) - |a|^4}\,; \tag{6}$$

for non-collinear digits, ψ is shifted by $\frac{1}{2}\arg\tau$. In general, the aspect ratio depends on u, v, and the complex value of τ (not just $|\tau|$).

Proof. Let Z be distributed according to the invariant measure μ. The fixed-point equation

$$E[h(Z)] = \frac{1}{N}\Big(\sum_{k\in I} E\big[h\big(a(s_kZ + t_k)\big)\big] + \sum_{k\in J} E\big[h\big(a(s_k\overline{Z} + t_k)\big)\big]\Big) \tag{7}$$

applied to $h(z) = z$ gives $E[Z] = a\big(u'\,E[Z] + v'\,\overline{E[Z]} + \frac{1}{N}\sum t_k\big)$ where $u' = \frac{1}{N}\sum_{k\in I} s_k$, $v' = \frac{1}{N}\sum_{k\in J} s_k$; since $|a|(|u'| + |v'|) \le |a| < 1$ and $\sum t_k = 0$, the unique solution is $E[Z] = 0$.

Applying (7) to $h(z) = |z|^2$: since $E[Z] = 0$, the cross terms $2\,\mathrm{Re}(\overline{s_kZ}\,t_k)$ vanish in expectation, and $|s_kz|^2 = |z|^2$, giving

$$E[|Z|^2] = |a|^2\big(E[|Z|^2] + \sigma^2\big),$$

so $E[|Z|^2] = |a|^2\sigma^2/(1-|a|^2)$.

Applying (7) to $h(z) = z^2$ and writing $\omega := E[Z^2]$: $(s_k z)^2 = s_k^2 z^2$ for preserving maps and $s_k^2 \overline{z}^2$ for reversing ones, while the cross terms $2 s_k z \cdot t_k$ vanish in expectation and the translations contribute $\sum t_k^2 = N\tau$, so

$$\omega = a^2 \big(u\,\omega + v\,\overline{\omega} + \tau \big). \tag{8}$$

Rewriting as the $\mathbb{R}$-linear system $(1 - a^2 u)\,\omega - a^2 v\,\overline{\omega} = a^2 \tau$ and applying Cramer's rule ($\Delta := |1 - a^2 u|^2 - |a|^4 |v|^2 > 0$ since $|a|^2(|u| + |v|) \le |a|^2 < 1$) gives (1). The ellipticity $\rho = |\omega| / E[|Z|^2]$ gives (2), and $\mathrm{AR}^2 = (1-\rho)/(1+\rho)$. The orientation (3) follows from the covariance decomposition in Remark 2.

When $v = 0$, the equation is $\mathbb{C}$-linear: $\omega = a^2 \tau / (1 - a^2 u)$, giving (4). When also $s_k = 1$ for all k, $u = 1$ and $\rho = (1 - |a|^2)\kappa / |1 - a^2|$, giving (5). Properties (a)–(c) follow: ρ is increasing in κ, vanishes at $\kappa = 0$, and $\rho = 1$ (i.e. $\mathrm{AR} = 0$) requires $\kappa = 1$ and $|1 - a^2| = 1 - |a|^2$, which holds iff $a \in \mathbb{R}$. □

When the contraction ratio is the reciprocal of a Gaussian integer and the translations are collinear ($\kappa = 1$), the aspect ratio is a quadratic irrational whose field is determined by a single Gaussian norm.

Corollary 2. *Let $\lambda \in \mathbb{Z}[i]$ with $|\lambda| > 1$ and $\lambda \notin \mathbb{R}$, and set $a = 1/\lambda$. For any IFS with $\kappa = 1$, the aspect ratio (5) lies in $\mathbb{Q}(\sqrt{d}\,)$, where d is the square-free part of $\mathbf{N}(\lambda^2 - 1)$. Explicitly,*

$$\mathrm{AR}^2 = \frac{\sqrt{\mathbf{N}(\lambda^2 - 1)} - (\mathbf{N}(\lambda) - 1)}{\sqrt{\mathbf{N}(\lambda^2 - 1)} + (\mathbf{N}(\lambda) - 1)}. \tag{9}$$

Proof. Substitute $|a|^2 = 1/\mathbf{N}(\lambda)$, $|1 - a^2| = \sqrt{\mathbf{N}(\lambda^2 - 1)}/\mathbf{N}(\lambda)$, and $\kappa = 1$ into (5). □

Table 1. Aspect ratios for small Gaussian integers λ, illustrating Corollary 2. The quadratic field $\mathbb{Q}(\sqrt{d}\,)$ is determined by the square-free part of $\mathbf{N}(\lambda^2 - 1)$.

λ	$\lambda^2 - 1$	$\mathbf{N}(\lambda^2 - 1)$	d	AR
$1 + i$	$-1 + 2i$	5	5	$1/\varphi = (\sqrt{5} - 1)/2 \approx 0.618$
$2 + i$	$2 + 4i$	20	5	$1/\varphi^3 = \sqrt{5} - 2 \approx 0.236$
$1 + 2i$	$-4 + 4i$	32	2	$\sqrt{2} - 1 \approx 0.414$

The first two rows of Table 1 share the same quadratic field $\mathbb{Q}(\sqrt{5})$ because the rational prime 5 divides both norms. This is not a coincidence: the Cayley-type transform (9) maps $\sqrt{\mathbf{N}(\lambda^2 - 1)}$ to an element of the real quadratic field $\mathbb{Q}(\sqrt{d}\,)$, and the particular metallic ratio that appears is controlled by the arithmetic of this field. In the first row, $\sqrt{5}$ yields a rational expression in φ thanks to the identity $\varphi^2 = \varphi + 1$: the golden ratio is the fundamental unit of the ring $\mathbb{Z}[\varphi]$, so every element of $\mathbb{Q}(\sqrt{5})$ in $(0, 1)$ of the form $(\sqrt{5} - n)/(\sqrt{5} + n)$ is a power of $1/\varphi$. The third row gives the *silver ratio* $\delta_S = \sqrt{2} - 1$ (fundamental unit of $\mathbb{Z}[\sqrt{2}]$).

Corollary 2 assumes $\kappa = 1$. The following extension covers arbitrary κ.

Corollary 3. *For any IFS $\{f_k(z) = a(z + t_k)\}$ with $a = 1/\lambda$, $\lambda \in \mathbb{Z}[i] \setminus \mathbb{R}$, and digits $t_k \in \mathbb{Z}[i]$, the anisotropy satisfies $\kappa^2 \in \mathbb{Q}$, and AR^2 lies in $\mathbb{Q}(\sqrt{d}\,)$ where d is the square-free part of $\mathbf{N}(\lambda^2 - 1) \cdot \big(\mathbf{N}(\lambda^2 - 1) - \kappa^2 (\mathbf{N}(\lambda) - 1)^2\big)$. In particular, the quadratic field of* AR *depends only on λ and κ^2, not on the individual digits.*

Proof. Write $\tau = \sum (t_k - \bar{t})^2 \in \mathbb{Z}[i]$ and $\sigma^2 = \sum |t_k - \bar{t}|^2 \in \mathbb{Z}$. Then $\kappa^2 = |\tau|^2/\sigma^4 = \mathbf{N}(\tau)/\sigma^4$, which is rational since $\mathbf{N}(\tau) \in \mathbb{Z}$. Substituting into (5) gives

$$\mathrm{AR}^2 = \frac{|1-a^2| - (1-|a|^2)\kappa}{|1-a^2| + (1-|a|^2)\kappa} = \frac{\sqrt{R} - (N-1)\kappa}{\sqrt{R} + (N-1)\kappa},$$

where $R = \mathbf{N}(\lambda^2 - 1)/N^2$. Clearing N^2: the numerator and denominator are $\sqrt{\mathbf{N}(\lambda^2-1)} \mp (N-1)\kappa$. Rationalising, $\mathrm{AR}^2 = 1 - 2(N-1)\kappa/\big(\sqrt{\mathbf{N}(\lambda^2-1)} + (N-1)\kappa\big)$, and since $(N-1)\kappa$ is $\sqrt{\mathbf{N}(\tau)}/\sigma^2 \cdot (N-1)/N$ — a rational multiple of $\sqrt{\mathbf{N}(\tau)}$, while $\sqrt{\mathbf{N}(\lambda^2-1)}$ is algebraic of degree ≤ 2, AR^2 lies in a quadratic extension of $\mathbb{Q}$. □

Remark 1. *For collinear translations ($\kappa = 1$), formula* (5) *shows that* AR *depends on a only through $(|a|^2, |1-a^2|)$: the attractor is nearly round when $|1-a^2|$ is large relative to $1-|a|^2$, and degenerates to a segment when $|1-a^2| \approx 1-|a|^2$ (i.e. a nearly real). For every a in the family, $\mathrm{AR}(a) \leq |a|$, with equality if and only if a is purely imaginary. Indeed, $|1-a^2| \leq 1+|a|^2$ (triangle inequality), with equality iff $a^2/|a^2| = -1$, i.e. $a \in i\mathbb{R}$. Substituting $|1-a^2| = 1+|a|^2$ into* (5) *gives $\mathrm{AR}^2 = |a|^2$. Figure 1 shows* AR *over the entire sub-family $|a| = 1/\sqrt{2}$.*

Remark 2. *The orientation formula* (3) *follows from the decomposition of the covariance matrix. Writing $\omega = |\omega|\, e^{2i\psi}$ for $Z = X + iY$ with $E[Z] = 0$:*

$$M = \frac{E[|Z|^2]}{2}\, I + \frac{|\omega|}{2} \begin{pmatrix} \cos 2\psi & \sin 2\psi \\ \sin 2\psi & -\cos 2\psi \end{pmatrix},$$

whose larger eigenvalue has eigenvector $(\cos\psi, \sin\psi)$. For the collinear case (6), $\mathrm{Im}(a^2)$ *measures the rotation per level of the IFS hierarchy, and $\mathrm{Re}(a^2) - |a|^4$ is the real drift; their ratio determines the principal-axis direction.*

3. The $\lambda_{N,\ell}$ family

A natural two-parameter family of plane-filling tiles arises by applying Theorem 1 to collinear digits. For integers $N \geq 2$ and $0 \leq \ell < 2\sqrt{N}$, let $\lambda_{N,\ell}$ be a root of $x^2 - \ell x + N$ (so $|\lambda_{N,\ell}|^2 = N$ and $\lambda_{N,\ell} \notin \mathbb{R}$). The digits $\{0, 1, \ldots, N-1\}$ form a complete residue system (CRS) of $\mathbb{Z}[\lambda_{N,\ell}]$ modulo $\lambda_{N,\ell}$. Indeed, every element $a + b\lambda_{N,\ell} \in \mathbb{Z}[\lambda_{N,\ell}]$ satisfies $\lambda_{N,\ell}(a + b\lambda_{N,\ell}) = a\lambda_{N,\ell} + b(\ell\lambda_{N,\ell} - N)$ (using $\lambda_{N,\ell}^2 = \ell\lambda_{N,\ell} - N$), whose integer part is $-bN$; hence $a + b\lambda_{N,\ell} \equiv 0 \pmod{\lambda_{N,\ell}}$ forces $b = 0$ and $N \mid a$, so $\lambda_{N,\ell}\mathbb{Z}[\lambda_{N,\ell}] \cap \mathbb{Z} = N\mathbb{Z}$ and the N residues $\{0, \ldots, N-1\}$ are distinct. The N-map IFS with $a = 1/\lambda_{N,\ell}$ and digits centred to $\sum t_j = 0$ therefore produces a tile. The aspect ratio and orientation of these tiles admit explicit formulas.

Proposition 4. *Let $N \geq 2$ and $0 \leq \ell < 2\sqrt{N}$, and let $\lambda = \lambda_{N,\ell}$ be a root of $x^2 - \ell x + N$. For the N-map tiling IFS with contraction $a = 1/\lambda$:*

(i) *$|\lambda^2 - 1|^2 = (N+1)^2 - \ell^2$, and the aspect ratio is*

$$\mathrm{AR}_{N,\ell} = \sqrt{\frac{\sqrt{(N+1)^2 - \ell^2} - (N-1)}{\sqrt{(N+1)^2 - \ell^2} + (N-1)}}. \tag{10}$$

(ii) *The major axis of inertia makes angle*

$$\psi_{N,\ell} = -\tfrac{1}{2}\arg(\lambda^2 - 1) \tag{11}$$

with the positive x-axis, where $\lambda^2 - 1 = \frac{1}{2}\big((\ell^2 - 2N - 2) + i\ell\sqrt{4N - \ell^2}\big)$.

(iii) *When* $\ell = 2$ *(the sub-family* $\lambda_N = 1 + i\sqrt{N-1}$*),* (10) *simplifies to* $\mathrm{AR}_N = (\sqrt{N+3} - \sqrt{N-1})/2$*, and* $\tan\psi$ *satisfies* $t^2 + t\sqrt{N-1} - 1 = 0$*. When additionally* $N-1 = m^2$*, this is the* m*-th metallic-ratio equation:* $m = 1$ *gives* φ*,* $m = 2$ *gives* $1 + \sqrt{2}$*.*

Proof. Part (i): Expanding $\lambda = (\ell + i\sqrt{4N - \ell^2})/2$ gives

$$|\lambda^2 - 1|^2 = \tfrac{1}{4}\big[(\ell^2 - 2N - 2)^2 + \ell^2(4N - \ell^2)\big] = (N+1)^2 - \ell^2.$$

Substituting $|a|^2 = 1/N$ and $|1 - a^2| = \sqrt{(N+1)^2 - \ell^2}/N$ into (5) yields (10). Part (ii): for collinear digits $\omega = a^2\sigma^2/(1 - a^2) = \sigma^2/(\lambda^2 - 1)$, so (3) gives $\psi = -\frac{1}{2}\arg(\lambda^2 - 1)$. Part (iii): Setting $\ell = 2$ gives $(N+1)^2 - 4 = (N-1)(N+3)$; the AR formula rationalises to $(\sqrt{N+3} - \sqrt{N-1})/2$. For the tangent, $\lambda^2 - 1 = (1 - N) + 2i\sqrt{N-1}$; the double-angle identity $\tan 2\psi = 2\tan\psi/(1 - \tan^2\psi) = 2/\sqrt{N-1}$ yields the stated equation. □

Selected values:

N	ℓ	$\lambda_{N,\ell}$	field	$\mathrm{AR}_{N,\ell}$
2	0	$i\sqrt{2}$	$\mathbb{Q}$	$1/\sqrt{2} \approx 0.707$ (rectangle)
2	1	$(1 + i\sqrt{7})/2$	$\mathbb{Q}(\sqrt{2})$	≈ 0.691 (tame twindragon)
2	2	$1 + i$	$\mathbb{Q}(\sqrt{5})$	$1/\varphi \approx 0.618$
5	2	$1 + 2i$	$\mathbb{Q}(\sqrt{2})$	$\sqrt{2} - 1 \approx 0.414$
5	4	$2 + i$	$\mathbb{Q}(\sqrt{5})$	$1/\varphi^3 \approx 0.236$

The "field" column records $\mathbb{Q}(\sqrt{d}\,)$ where d is the square-free part of $(N+1)^2 - \ell^2$. The entry $(N, \ell) = (2, 1)$ recovers the tame twindragon (see below), confirming that it belongs to the same two-parameter family. The entry $(N, \ell) = (5, 4)$ gives $\lambda_{5,4} = 2 + i$ — the same contraction as the 2-map tile in row 2 of Table 1, illustrating again that AR depends only on $a = 1/\lambda$ (Theorem 1).

For fixed ℓ, $\mathrm{AR}_{N,\ell} \to 0$ as $N \to \infty$ (stronger contraction elongates the attractor). For fixed N, increasing ℓ toward $2\sqrt{N}$ drives a toward the real axis and $\mathrm{AR} \to 0$; setting $\ell = 0$ gives $\lambda = i\sqrt{N}$ (purely imaginary a) and $\mathrm{AR} = 1/\sqrt{N}$, the maximum for a given N. For $N = 2$ the IFS $\{(i/\sqrt{2})(z \pm 1)\}$ has attractor the rectangle $[-1, 1] \times [-\sqrt{2}, \sqrt{2}]$ (one verifies that f_1 and f_2 map it to the top and bottom halves); this rectangle tiles the plane by translation.

Tame twindragon. The entry $(N, \ell) = (2, 1)$ gives $\lambda = (1 + i\sqrt{7})/2$ and the attractor is the *tame twindragon* [1, p. 550], the attractor of $\{f_{1,2}(z) = a(z \pm 1)\}$ with $a = (1 - i\sqrt{7})/4$. One computes $1 - a^2 = (11 + i\sqrt{7})/8$, hence $|1 - a^2| = \sqrt{2}$, and Theorem 1 gives

$$\mathrm{AR}^2 = \frac{\sqrt{2} - \frac{1}{2}}{\sqrt{2} + \frac{1}{2}} = \frac{2\sqrt{2} - 1}{2\sqrt{2} + 1} = \frac{9 - 4\sqrt{2}}{7} \approx 0.478, \qquad \mathrm{AR} \approx 0.691.$$

This tile arises as the numeration digit set for the complex base $\beta = (1 + i\sqrt{7})/2$ (satisfying $\beta^2 - \beta + 2 = 0$ and $|\beta|^2 = 2$) via $a = 1/\beta$;[1] it is marked in Figure 1.

The connection to metallic ratios in part (iii) is not accidental. The m-th metallic ratio $\mu_m = (m + \sqrt{m^2 + 4})/2$ is the fundamental unit of $\mathbb{Z}[\mu_m] \subset \mathbb{Q}(\sqrt{m^2 + 4})$. In the $\ell = 2$ sub-family, $\sqrt{m^2 + 4} = \sqrt{N + 3}$ and the Cayley-type transform (9) maps $(N+1)^2 - 4 = (N-1)(N+3)$ into $\mathbb{Q}(\sqrt{N+3})$. For $N = 2$ and $N = 5$ the aspect

[1] The attractor is the set $\{\sum_{k \geq 1} d_k \beta^{-k} : d_k \in \{-1, +1\}\} \subset \mathbb{C}$.

ratio is an exact power of the fundamental unit ($1/\varphi$ and $\sqrt{2}-1$ respectively); for other N it remains in $\mathbb{Q}(\sqrt{N+3})$ but is not generally a unit.

Remark 3. *Both the aspect ratio and the principal-axis orientation of the $\lambda_{N,\ell}$-family are governed by a single element of the ring of integers: $\lambda^2-1=\ell\lambda-(N{+}1)\in\mathbb{Z}[\lambda]$. Its norm $(N{+}1)^2-\ell^2$ determines* AR *via* (10)*, and its argument determines ψ via* (11)*. This admits a hierarchical interpretation. The random-iteration representation $Z=\sum_{n\geq 1}a^n\varepsilon_n$ (ε_n i.i.d. from the centred digits) decomposes $E[Z^2]$ as $\sum_{n\geq 1}\sigma^2a^{2n}$: each level n of the IFS contributes σ^2/N^n to the anisotropy along direction $2n\arg(a)=-2n\arg(\lambda)$. The geometric sum of these rotated contributions points in the direction $\arg\big(a^2/(1-a^2)\big)=-\arg(\lambda^2-1)$, which is precisely twice the principal-axis angle. At the boundary $\ell=0$ the vector $\lambda^2-1=-(N{+}1)$ is real negative, giving $\psi=-\pi/2$ (vertical major axis); as $\ell\to 2\sqrt{N}$ the vector tends to $2(N-1)>0$, giving $\psi\to 0$ (horizontal, degenerate).*

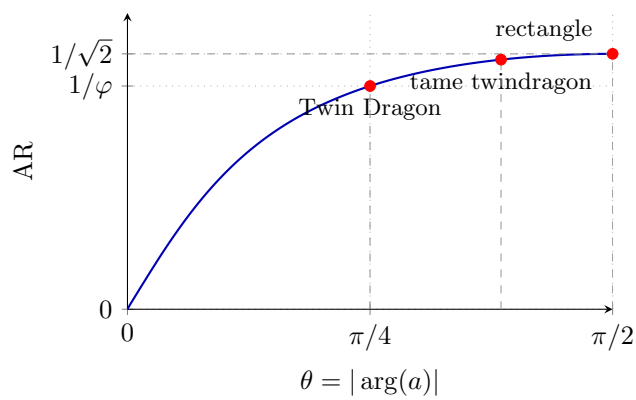

FIGURE 1. Aspect ratio $\mathrm{AR}(\theta)$ of the attractor of $\{a(z\pm 1)\}$ for $|a|=1/\sqrt{2}$, as a function of $\theta=|\arg(a)|$. The Twin Dragon ($\theta=\pi/4$), the tame twindragon ($\theta=\arctan\sqrt{7}$), and the rectangle attractor ($\theta=\pi/2$) are marked in red.

4. Tiles over $\mathbb{Z}[i]$

In this section we specialise to tiles over $\mathbb{Z}[i]$: the contraction is $a=1/\lambda$ for a Gaussian integer λ and the translations form a CRS modulo λ. Theorem 1 then applies with $|a|^2=1/N$ and κ determined by the choice of CRS.

The dependence on κ is already visible for a single expansion factor. Take $\lambda=2+i$ ($N{=}5$). The collinear digit set $\{0,1,2,3,4\}$ gives $\kappa=1$ and $\mathrm{AR}=1/\varphi^3$ (row $(5,4)$ of the table in Section 3). Replacing it by the two-dimensional CRS $\{0,\ 1,\ -i,\ i,\ 1{+}i\}$ (which is still a complete residue system of $\mathbb{Z}[i]$ modulo $2{+}i$, since $i\equiv 3 \pmod{2+i}$) yields $\sigma^2=4/5$, $\tau=(-8+6i)/25$, and $\kappa=1/2$. Substituting into (5):

$$\mathrm{AR}^2=\frac{\sqrt{5}-1}{\sqrt{5}+1}=\frac{3-\sqrt{5}}{2}=\frac{1}{\varphi^2},\qquad \mathrm{AR}=\frac{1}{\varphi}.$$

Thus the same golden ratio governs this 5-map tile with $\kappa=1/2$ as it does the Twin Dragon ($N{=}2$, $\kappa{=}1$), while collinear digits for the same λ give the smaller

value $1/\varphi^3$ (Figure 2). A natural question arises: for which κ does the aspect ratio equal $1/\mu_m$, the reciprocal of the m-th metallic mean?

Proposition 5. *Let $\lambda = p+qi \in \mathbb{Z}[i]\setminus\mathbb{R}$ with $|\lambda|^2 = N$, and let $m \geq 1$ be a positive integer. The digit anisotropy required for $\mathrm{AR} = 1/\mu_m$ (the reciprocal m-th metallic mean $\mu_m = (m+\sqrt{m^2+4})/2$) is*

$$\kappa_m^* = \frac{m\,|\lambda^2-1|}{\sqrt{m^2+4}\,(N-1)}, \tag{12}$$

and $\kappa_m^* \leq 1$ if and only if $mq \leq N-1$. In particular:

(a) *For $m = 1$ (golden ratio), $\kappa_1^* \leq 1$ for every λ, with equality only for $\lambda = 1+i$.*
(b) *The collinear case $\kappa = 1$ corresponds to $m = (N-1)/q$, provided this is an integer; thus the collinear aspect ratio of Corollary 2 equals $1/\mu_{(N-1)/q}$.*
(c) *For $q = 1$ (i.e. $\lambda = p+i$), every metallic ratio $\mu_1, \ldots, \mu_{p^2}$ is accessible.*

Proof. Setting $\mathrm{AR}^2 = 1/\mu_m^2$ in (5) and solving for κ gives $\kappa = |1-a^2|\cdot m/\big((1-|a|^2)\sqrt{m^2+4}\big)$; substituting $a = 1/\lambda$, $|a|^2 = 1/N$, $|1-a^2| = \sqrt{\mathbf{N}(\lambda^2-1)}/N$ yields (12).

The condition $\kappa_m^* \leq 1$ becomes $m^2|\lambda^2-1|^2 \leq (m^2+4)(N-1)^2$. Since $|\lambda^2-1|^2 = (N-1)^2+4q^2$, this simplifies to $4m^2q^2 \leq 4(N-1)^2$, i.e. $mq \leq N-1$.

(a): $q \leq N-1$ holds since $N = p^2+q^2 \geq q+1$ for $p \geq 1$. Equality $q = N-1$ gives $p^2 = 1-q(q-1)$; for $q = 1$ this yields $p = 1$, $\lambda = 1+i$; for $q \geq 2$ one has $p^2 < 0$, which is impossible.

(b): $\kappa_m^* = 1$ iff $m^2q^2 = (N-1)^2$, so $m = (N-1)/q$.

(c): $q = 1$ gives $mq = m \leq N-1 = p^2$. □

Corollary 6. *Up to associates, $\kappa_1^* = 1/2$ if and only if $\lambda \in \{2+i,\ 2+3i,\ 1+4i\}$ ($N = 5, 13, 17$). More generally, $\kappa_1^* = 1/m$ for an integer $m \geq 1$ only for $m = 1$ ($\lambda = 1+i$) and $m = 2$ (the three primes above).*

Proof. Write $\lambda = p+qi$, $N = p^2+q^2$. Setting $\kappa_1^* = 1/m$ in $|\lambda^2-1|^2 = 5(N-1)^2/m^2$ and using $|\lambda^2-1|^2 = (N-1)^2+4q^2$ gives

$$4q^2 = (N-1)^2\big(\tfrac{5}{m^2}-1\big).$$

For $m = 1$ this is $4q^2 = 4(N-1)^2$, so $q = N-1 = 1$, $N = 2$: only $\lambda = 1+i$. For $m = 2$: $4q^2 = \frac{1}{4}(N-1)^2$, so $N = 4q+1$, $p^2 = 4q+1-q^2$, and the positivity condition $p^2 \geq 1$ becomes $p^2+(q-2)^2 = 5$, i.e. a representation of 5 as a sum of two squares. The three solutions with $q \geq 1$, $p \geq 1$ are $(p,q) = (2,1), (2,3), (1,4)$. For $m \geq 3$: $5/m^2-1 < 0$, so $4q^2 < 0$ — impossible. □

Theorem 7. *For every integer $m \geq 1$:*

(a) *There exists a self-similar plane-filling tile in $\mathbb{R}^2$ whose aspect ratio equals $1/\mu_m$, the reciprocal of the m-th metallic mean.*
(b) *Any such tile over $\mathbb{Z}[i]$ requires at least $N_{\min}(m) = \lceil\sqrt{m}\,\rceil^2+1$ maps.*
(c) *When m is a perfect square, $N_{\min}(m) = m+1$, achieved by the collinear CRS for $\lambda = \sqrt{m}+i$.*

Proof. (a): Let $\lambda = 1+mi \in \mathbb{Z}[i]$. Then $N = |\lambda|^2 = m^2+1$ and $\mathrm{Im}(\lambda) = m$. The standard digits $\{0, 1, \ldots, m^2\}$ form a CRS modulo λ (§3); the induced N-map IFS satisfies the open set condition [1], so its attractor is a plane-filling tile. The collinear

digit set has $\kappa = 1$, and Proposition 5(b) gives $m = (N-1)/\operatorname{Im}(\lambda) = m^2/m$, hence $\mathrm{AR} = 1/\mu_m$.

(b): For any $\lambda = p+qi$ with $mq \le N-1$ (Proposition 5), we have $N = p^2+q^2 \ge q(m-q)+1+q^2 = qm+1$, which is minimised at $q=1$, giving $N \ge \lceil\sqrt{m}\,\rceil^2+1$.

(c): When $m = k^2$, the choice $\lambda = k+i$ yields $N = k^2+1 = m+1$ and the collinear CRS has $\kappa = 1$ with $(N-1)/q = m$. □

Remark 4. *Not every integer $N \ge 2$ is the norm of a Gaussian integer with non-zero imaginary part. By the sum-of-two-squares theorem, $N = p^2+q^2$ with $q \ne 0$ if and only if every prime factor of N congruent to $3 \pmod 4$ appears to an even power. The first admissible values are $N = 2, 4, 5, 8, 9, 10, 13, \ldots$; in particular $N = 3, 6, 7, 11, 14, 15$ are not representable.*

Remark 5. *Item (b) of Proposition 5 recovers every row of Table 1 as a special case: $\lambda = 1+i$ gives $m = 1$ and $\mathrm{AR} = 1/\mu_1 = 1/\varphi$; $\lambda = 2+i$ gives $m = 4$ and $\mathrm{AR} = 1/\mu_4 = 1/(2+\sqrt{5}) = \sqrt{5}-2 = 1/\varphi^3$; $\lambda = 1+2i$ gives $m = 2$ and $\mathrm{AR} = 1/\mu_2 = \sqrt{2}-1$. Thus the collinear aspect ratios of Section 2 and the golden-ratio tiles of Table 2 are part of a single metallic-ratio spectrum parametrised by κ. Table 2 lists CRS digit sets achieving κ_1^* (i.e. $\mathrm{AR} = 1/\varphi$) for small Gaussian primes λ.*

TABLE 2. Complete residue systems $D \subset \mathbb{Z}[i]$ (modulo λ, centred to $\sum t_k = 0$) achieving $\mathrm{AR} = 1/\varphi$.

λ	N	κ^*	*digit set D*
$1+i$	2	1	$\{0, \pm 1\}$
$2+i$	5	$1/2$	$\{0, 1, -i, i, 1+i\}$
$1+2i$	5	$\sqrt{2/5}$	$\{-1-2i, -1, i, 1, 1+i\}$
$1+3i$	10	$\sqrt{13/45}$	$\{-2, -2+2i, -1-i, -1, -1+i, -i, 0, i, 1-2i, 1\}$
$2+3i$	13	$1/2$	$\{3-2i, 5i, 3-5i, 3, \ldots, 12\}$
$1+4i$	17	$1/2$	$\{-3, \ldots, 2+2i\}$ *(17 digits)*

Rows 3 and 4 have irrational κ^* ($\sqrt{2/5}$ and $\sqrt{13/45}$), yet Gaussian-integer digits realise these values exactly. Corollary 6 explains why rows 2, 5, and 6 all share $\kappa^* = 1/2$: only three Gaussian primes attain this value, and the proof reduces to the unique representation $5 = 1^2+2^2$.

Proposition 8. *Let $\lambda \in \mathbb{C} \setminus \mathbb{R}$ with $|\lambda|^2 = N \ge 2$. Then*

$$|\lambda^2-1|^2 = (N-1)^2 + 4\,(\operatorname{Im}\lambda)^2, \tag{13}$$

and the digit anisotropy (12) satisfies $\kappa_m^* \le 1$ if and only if $m \le (N-1)/|\operatorname{Im}\lambda|$. The collinear case $\kappa = 1$ corresponds to $m = (N-1)/|\operatorname{Im}\lambda|$, which is a positive integer only when $|\operatorname{Im}\lambda|$ divides $N-1$.

Proof. Write $\lambda = x+iy$, so $N = x^2+y^2$ and $\lambda^2-1 = (x^2-y^2-1)+2ixy$. Then

$$|\lambda^2-1|^2 = (x^2-y^2-1)^2+4x^2y^2 = (x^2+y^2-1)^2+4y^2 = (N-1)^2+4y^2,$$

since, by direct expansion, $(x^2-y^2-1)^2+4x^2y^2-(x^2+y^2-1)^2 = 4y^2$. Substituting into (12): $\kappa_m^* \le 1$ iff $m^2\big((N-1)^2+4y^2\big) \le (m^2+4)(N-1)^2$, i.e. $4m^2y^2 \le 4(N-1)^2$, so $m \le (N-1)/|y|$. □

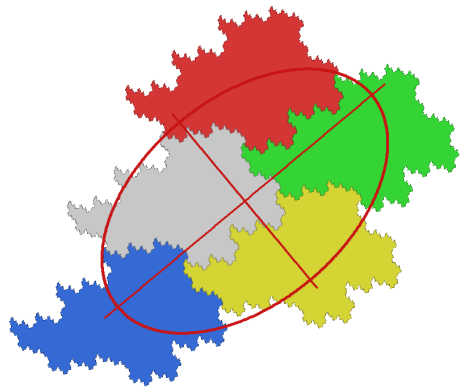

FIGURE 2. The 5-map tile with $\lambda = 2 + i$ and digit set $\{0, 1, -i, i, 1+i\}$ ($\kappa = 1/2$). Its 1.5-sigma principal-axis ellipse (red) confirms AR $= 1/\varphi \approx 0.618$, the same value as the Twin Dragon (Figure 3).

Corollary 9. *Among all imaginary quadratic integer rings, $\mathbb{Z}[i]$ is the unique one where collinear self-similar tiles can have metallic-ratio aspect ratios.*

Proof. For $\lambda = p + q\sqrt{-k} \in \mathbb{Z}[\sqrt{-k}]$ ($k \geq 2$, square-free, $q \neq 0$), $\operatorname{Im}\lambda = q\sqrt{k} \notin \mathbb{Q}$, so $(N-1)/\operatorname{Im}\lambda$ is irrational. The same holds for the full ring of integers of $\mathbb{Q}(\sqrt{-3})$: if $\lambda = a + b\omega \in \mathbb{Z}[\omega]$ with $\omega = e^{2\pi i/3}$ and $b \neq 0$, then $\operatorname{Im}\lambda = b\sqrt{3}/2 \notin \mathbb{Q}$. In $\mathbb{Z}[i]$, however, $\operatorname{Im}\lambda = q \in \mathbb{Z}$, and taking $q = 1$ gives $m = (N-1)/1 = p^2$, always a positive integer. □

Remark 6. *Proposition 8 shows that the identity $|\lambda^2 - 1|^2 - (N-1)^2$ is controlled entirely by $\operatorname{Im}\lambda$. Specialising to each ring: $\mathbb{Z}[i]$: $4q^2$; $\mathbb{Z}[\omega]$: $3b^2$; $\mathbb{Z}[\sqrt{-k}]$: $4kq^2$. The upper bound $m_{\max} = (N-1)/|\operatorname{Im}\lambda|$ is rational only for $\mathbb{Z}[i]$. Nonetheless, by choosing $\kappa < 1$ (non-collinear digits), $\kappa_m^* \leq 1$ is guaranteed for every imaginary quadratic ring whenever N is large enough. A computational search over $\mathbb{Z}[\omega]$ confirms that complete residue systems achieving $\kappa = \kappa_1^*$ (i.e. AR $= 1/\varphi$) exist for all Eisenstein primes of norm $N \geq 7$. However, for $N = 3$ ($\lambda = 1 - \omega$, $\kappa_1^* = \sqrt{7/20}$), an exhaustive search over shifts $|c_k| \leq 8$ fails to achieve κ_1^* (best approximation $\approx 3 \times 10^{-5}$), suggesting that three digits provide too few degrees of freedom in the hexagonal lattice.*

Remark 7. *For* collinear *digits ($\kappa = 1$) the condition $m = (N-1)/q$ (Proposition 5(b)) requires $p^2 = q(m-q) + 1$ to be a perfect square. Taking $q = m$ gives the universal upper bound $N_{\mathrm{col}}(m) \leq m^2 + 1$, but smaller q may yield a smaller N. For $m = k^2$ a perfect square, $q = 1$ gives $N_{\mathrm{col}}(k^2) = k^2 + 1 = N_{\min}(k^2)$, so the collinear and non-collinear minima coincide. In contrast, for non-square m a gap opens: e.g. $N_{\min}(5) = 10$ while $N_{\mathrm{col}}(5) = 26$, and $N_{\min}(7) = 10$ while $N_{\mathrm{col}}(7) = 50$. In general, $N_{\mathrm{col}}(m)$ equals the minimum of $q^2 + q(m-q) + 1$ over those $q \in \{1, \ldots, m\}$ for which $q(m-q) + 1$ is a perfect square — a Diophantine condition whose behaviour mirrors the distribution of square values of binary quadratic forms.*

5. The Twin Dragon

The Twin Dragon $A \subset \mathbb{C}$ is the attractor of the IFS [4]

$$A = f_1(A) \cup f_2(A), \qquad f_{1,2}(z) = \frac{1-i}{2}(z \pm 1). \tag{14}$$

Both maps have the same contraction ratio $r = |(1-i)/2| = 1/\sqrt{2}$, so the similarity dimension is $\alpha = -\log 2/\log r = 2$. The open set condition holds [1, Corollary 3]: the Twin Dragon is a digit tile for $\mathbb{Z}[i]/(1+i)\mathbb{Z}[i]$, so its translates by $\mathbb{Z}[i]$ tile $\mathbb{C}$ with the interiors pairwise disjoint. By [6, §5] the self-similar measure therefore coincides with $\mathcal{H}^\alpha|_A$. Since $\alpha = 2$ equals the ambient dimension, $\mathcal{H}^2$ is Lebesgue area, and the covariance matrix M describes the actual geometric shape (Figure 3).

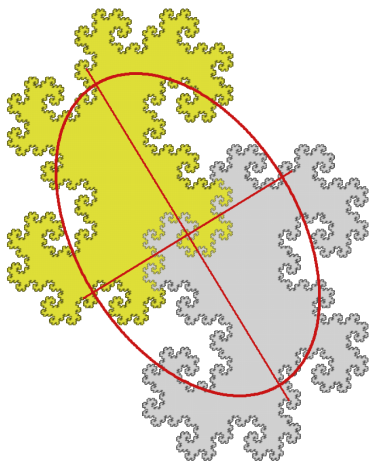

Figure 3. The Twin Dragon with its 1.5-sigma principal-axis ellipse (red). The major axis lies along $(1, -\varphi)$ and the minor axis along $(\varphi, 1)$; their ratio of lengths equals the aspect ratio $\mathrm{AR} = 1/\varphi \approx 0.618$.

Substituting $a = (1-i)/2$ into Theorem 1 (with $N = 2$, $t_k = \pm 1$, so $\sigma^2 = 1$): $|a|^2 = \frac{1}{2}$, $a^2 = -\frac{i}{2}$, $1 - a^2 = \frac{2+i}{2}$, so

$$E[Z^2] = \frac{a^2}{1-a^2} = \frac{-i}{2+i} = \frac{-1-2i}{5}, \qquad E[|Z|^2] = \frac{|a|^2}{1-|a|^2} = 1.$$

The covariance matrix follows via $M_{11} = \frac{1}{2}(E[|Z|^2] + \mathrm{Re}\, E[Z^2])$, $M_{22} = \frac{1}{2}(E[|Z|^2] - \mathrm{Re}\, E[Z^2])$, $M_{12} = \frac{1}{2}\mathrm{Im}\, E[Z^2]$:

$$M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}. \tag{15}$$

The matrix M has $\mathrm{tr}\, M = 1$ and $\det M = \frac{1}{5}$, so its eigenvalues are

$$I_{1,2} = \frac{1 \mp \frac{1}{\sqrt{5}}}{2}. \tag{16}$$

The geometric aspect ratio — the ratio of standard deviations along the principal axes of M — is

$$\sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{\sqrt{5}-1}{\sqrt{5}+1}} = \sqrt{\frac{3-\sqrt{5}}{2}} = \frac{1}{\varphi}, \tag{17}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.

Corollary 10. *The attractor of the IFS* $\{f_{1,2}(z) = \frac{1-i}{2}(z \pm 1)\}$ *has covariance matrix* $M = \frac{1}{5}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 3 \end{smallmatrix}\right)$*, geometric aspect ratio* $1/\varphi = (\sqrt{5}-1)/2 \approx 0.618$*, and major-axis direction* $(1, -\varphi)$ *(angle* $-\arctan\varphi \approx -58.3^\circ$ *with the horizontal), where* $\varphi = (1+\sqrt{5})/2$ *is the golden ratio.*

Proof. This is the case $\lambda = 1+i$ of Corollary 2 ($\mathbf{N}(\lambda^2-1) = 5$, $d = 5$), with the orientation from Proposition 4(ii). □

Remark 8. *The Twin Dragon is the* $(N, \ell) = (2, 2)$ *instance of Proposition 4, giving* $\tan\psi = -\varphi$ *(i.e.* $\psi \approx -58.3^\circ$*). Thus the same constant* φ *governs both the aspect ratio (*$\mathrm{AR} = 1/\varphi$*) and the orientation (*$\tan\psi = -\varphi$*).*

Remark 9. *The appearance of* φ *is unexpected: the Twin Dragon is defined by the Gaussian integer* $1+i$*, with no pentagon, decagon, or Fibonacci recurrence in the construction. The number* 5 *— and hence* φ *— can be traced to a single Gaussian prime. By the ergodic theorem for random iteration* [2]*,* μ *is the distribution of the random series* $Z = \sum_{k\geq 1} \varepsilon_k a^k$ *with* $a = (1-i)/2$ *and* $\varepsilon_k \overset{\text{iid}}{\sim} \{\pm 1\}$*. Since* $1 - a^2 = (2+i)/2$ *and* $\kappa = 1$ *(real translations), the ellipticity is* $\rho = (1-|a|^2)\kappa/|1-a^2| = 1/\sqrt{5}$*, where* $5 = \mathbf{N}(2+i)$ *is the norm of the Gaussian prime* $2+i$*. Thus* φ *enters via the Gaussian factorisation* $5 = (2+i)(2-i)$ *in* $\mathbb{Z}[i]$*, and the identity* $\mathrm{AR}^2 = (1-\rho)/(1+\rho)$ *from* (5) *immediately gives* $1/\varphi^2$ *(using* $\varphi^2 = \varphi + 1$*).*

Further directions. An explicit closed-form construction of a CRS achieving $\kappa = \kappa_m^*$ for all N would be desirable; this amounts to a system of quadratic Diophantine equations in the digit shifts $c_k \in \mathbb{Z}[i]$ (where $d_k = k + c_k\lambda$), which does not appear to simplify into a universal formula.

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